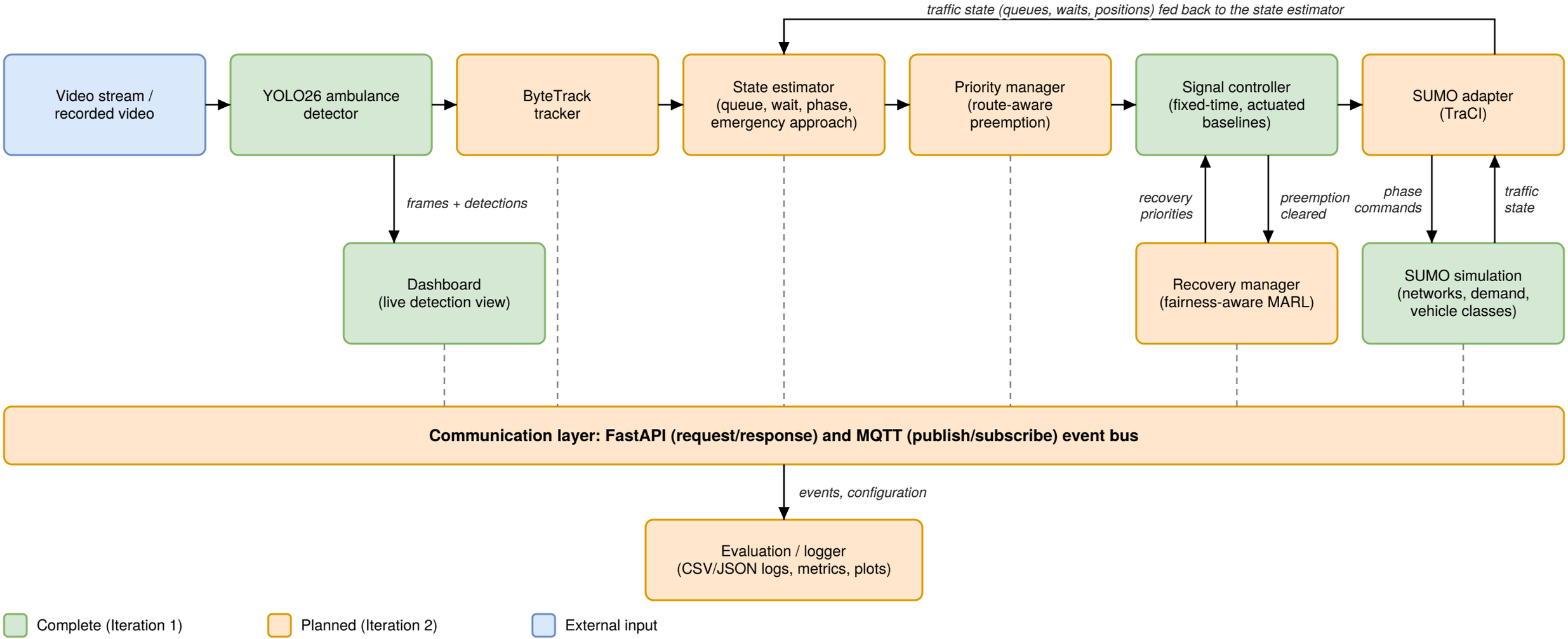

traffic state (queues, waits, positions) fed back to the state estimator
Video stream / recorded video
YOLO26 ambulance detector
ByteTrack tracker
State estimator (queue, wait, phase, emergency approach)
Priority manager (route-aware preemption)
Signal controller (fixed-time, actuated baselines)
SUMO adapter (TraCI)
frames + detections
Dashboard (live detection view)
recovery priorities
preemption cleared
phase commands
traffic state
Recovery manager (fairness-aware MARL)
SUMO simulation (networks, demand, vehicle classes)
Communication layer: FastAPI (request/response) and MQTT (publish/subscribe) event bus
events, configuration
Evaluation / logger (CSV/JSON logs, metrics, plots)
Complete (Iteration 1)
Planned (Iteration 2)
External input